

Advancements in Retrieval-Augmented Generation Systems

Retrieval-Augmented Generation (RAG) integrates retrieval mechanisms with LLMs.

Key Architecture: Parent-Child Chunking allows embedding small child chunks.

Hybrid Search combines Dense FAISS embeddings and Sparse BM25 lexical search.

Cross-Encoder Reranking scores candidate passages using full cross-attention.

Groq accelerators provide ultra-fast inference with sub-second response times.